BDA Practical No-5 using Windows Command Prompt

-By Aditya Nikam(BE AI & DS)

Step 1:-Open Ubuntu Terminal and type the command given below to verify whether Mongo DB is installed or not on the system

mongod –version

If Mongo DB is installed then the output will show you the version number of Mongo DB like shown below:-

```
C:\Users\HP>mongod --version
db version v8.3.1
Build Info: {
    "version": "8.3.1",
    "gitVersion": "094e631246d5b5bee5bd5f20b12f882bc8e286ea",
    "modules": [],
    "allocator": "tcmalloc-gperf",
    "environment": {
        "distmod": "windows",
        "distarch": "amd64",
        "target_arch": "amd64"
    }
}
```

If Mongo DB is not installed then refer my "Mongo DB Installation for Windows"pdf to install Mongo DB

Step 2:-Now type the following command:-

mongosh

You will output like shown below.Now You can type your Mongo DB commands:-

```
C:\Users\HP>mongosh
Current Mongosh Log ID: 6a0024a871b6cbc712abc113
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.8.3
Using MongoDB:          8.3.1
Using Mongosh:          2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-05-10T11:27:31.460+05:30: Access control is not enabled for the database. Read and write access to data and conf
iguration is unrestricted
------

test>
```

Step 3:-Now type the following command:-

use studentDB

```
test> use studentDB
switched to db studentDB
```

You can have any other name as well instead of "studentDB" for database

Step 4:-Now type:-

db.createCollection("students")

```
studentDB> db.createCollection("students")
{ ok: 1 }
```

You can have any other name as well instead of "students" for collection

Step 5:-Now type the following command:-

db.students.insertOne({rollno: 1, name: "Rutvik", marks: 80, city: "Pune"})

```
studentDB>  db.students.insertOne({rollno: 1, name: "Rutvik", marks: 80, city: "Pune"}
{
  acknowledged: true,
  insertedId: ObjectId('6a0012e5789e74b80e9df8b6')
}
```

Step 6:-Now type:-

db.students.insertMany([

{rollno: 2, name: "Aditya", marks: 85, city: "Nashik"},

{rollno: 3, name: "Priya", marks: 90, city: "Nashik"},

{rollno: 4, name: "Amit", marks: 75, city: "Mumbai"},

{rollno: 5, name: "Sneha", marks: 82, status: "pass", city: "Mumbai"}

])

```
studentDB> db.students.insertMany([
|     {rollno: 2, name: "Aditya", marks: 85, city: "Nashik"},
|     {rollno: 3, name: "Priya", marks: 90, city: "Nashik"},
|     {rollno: 4, name: "Amit", marks: 75, city: "Mumbai"},
|     {rollno: 5, name: "Sneha", marks: 82, status: "pass", city: "Mumbai"}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0012f1789e74b80e9df8b7'),
    '1': ObjectId('6a0012f1789e74b80e9df8b8'),
    '2': ObjectId('6a0012f1789e74b80e9df8b9'),
    '3': ObjectId('6a0012f1789e74b80e9df8ba')
  }
}
```

Step 7:-Now type:-

db.students.find().pretty()

```
[
  {
    _id: ObjectId('6a0012e5789e74b80e9df8b6'),
    rollno: 1,
    name: 'Rutvik',
    marks: 80,
    city: 'Pune'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 90,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b9'),
    rollno: 4,
    name: 'Amit',
    marks: 75,
    city: 'Mumbai'
  },
  {
```

Step 8:-Now type to print the students who marks are greater than 80:-

db.students.find({marks: {$gt: 80}}).pretty()

```
studentDB> db.students.find({marks: {$gt: 80}}).pretty()
[
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 90,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8ba'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  }
]
```

Step 9:-Find any student who lives in either Nashik OR Mumbai.

db.students.find({city: {$in: ["Nashik", "Mumbai"]}}).pretty()

```
studentDB>  db.students.find({city: {$in: ["Nashik", "Mumbai"]}}).pretty()
[
  {
    _id: ObjectId('6a001d62789e74b80e9df8c1'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a001d62789e74b80e9df8c2'),
    rollno: 3,
    name: 'Priya',
    marks: 90,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a001d62789e74b80e9df8c3'),
    rollno: 4,
    name: 'Amit',
    marks: 75,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('6a001d62789e74b80e9df8c4'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  }
]
```

Step 10:- Update Amit's marks to 80 using following command:-

db.students.updateOne({rollno: 3}, {$set: {marks: 80}})

```
studentDB> db.students.updateOne({rollno: 3}, {$set: {marks: 80}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Step 11:- If we want to find everyone but sort them by marks (Highest to Lowest). The '-1' means descending order. '1' would be ascending.

db.students.find().sort({marks: -1}).pretty()

```
[
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8ba'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  },
  {
    _id: ObjectId('6a0012e5789e74b80e9df8b6'),
    rollno: 1,
    name: 'Rutvik',
    marks: 80,
    city: 'Pune'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 80,
    city: 'Nashik'
  },
```

Step 12:- Update anyone with marks less than 65 to have a "fail" status using follwoing command

db.students.updateMany({marks: {$lt: 65}}, {$set: {status: "fail"}})

```
studentDB>  db.students.updateMany({marks: {$lt: 65}}, {$set: {status: "fail"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
```

Step 13:- Delete Rutvik (Roll No 1) from the database using the following command

db.students.deleteOne({rollno: 1})

```
studentDB>  db.students.deleteOne({rollno: 1})
{ acknowledged: true, deletedCount: 1 }
```

Step 14:-If instead of printing out all the data, if we just want exactly how many students are left in the database then use the following command.

db.students.countDocuments()

```
studentDB> db.students.countDocuments()
4
```

Step 15:-Print all the records

db.students.find().pretty()

```
[
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b7'),
    rollno: 2,
    name: 'Aditya',
    marks: 85,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b8'),
    rollno: 3,
    name: 'Priya',
    marks: 80,
    city: 'Nashik'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8b9'),
    rollno: 4,
    name: 'Amit',
    marks: 75,
    city: 'Mumbai'
  },
  {
    _id: ObjectId('6a0012f1789e74b80e9df8ba'),
    rollno: 5,
    name: 'Sneha',
    marks: 82,
    status: 'pass',
    city: 'Mumbai'
  }
```

Step 16:-Now stop Mongo DB service to prevent your laptop/computer from draining power or if you want to exit from Mongo DB shell then use Ctrl +C two times